

MODEL FITNESS

Churn Analysis

Data-driven retention strategy to maximize user lifetime value and ensure sustainable growth.



Introduction and Objectives



Identify Risk

Predict accurately which customers have a high probability of leaving the gym within the next month.



Analyze Factors

Discover the specific behaviors and socio-demographic characteristics that highly influence cancellation.



Design Strategies

Propose concrete, targeted retention actions based entirely on extracted data evidence.



Maximize Value

Optimize the overall user experience to guarantee a sustainable, long-term revenue growth trajectory.

Analysis Methodology



Exploration (EDA)

Comprehensive analysis of data distribution and comparison of averages to understand the gym's initial retention characteristics.



Predictive Modeling

Training and benchmarking of powerful classification algorithms (Logistic Regression vs. Random Forest) to predict churn.



Segmentation

Application of advanced clustering techniques with K-Means to group users into specific profiles and analyze cancellation rates.

Dataset Overview

4,000

Records Analyzed

26.5%

Global Churn Rate

The processed dataset contains detailed demographic information, recent usage behavior, contract type, and consumption level of additional services. There were **no null values**, allowing for a structured and comprehensive analysis of Model Fitness's customer loyalty.

Retained vs. Churned

Retained Customers

- > Extended contracts (average duration of ~5.7 months).
- > High frequency of recurring visits (~2.0 per week).
- > Strong participation and consistency in group classes.
- > Higher spending volume on additional services.

Churned Customers

- > Extremely short contracts (average ~1.7 months).
- > Low recent attendance frequency (~1.0 per week).
- > Shorter lifetime and minimal interaction.
- > Minimal level of complementary spending at the facilities.

Determining Factors

Variable Correlation

Statistical analysis confirms that **usage frequency in the last month** is the single most powerful indicator for predicting whether a customer will imminently leave the gym.

Additionally, formal **contract duration** and **lifetime** act as critical mitigating elements, strongly reducing the churn rate.

Other demographic variables, such as gender or whether the user provides a contact number, show minimal predictive impact in this model.



Predictive Model Performance

EVALUATION METRIC	LOGISTIC REGRESSION	RANDOM FOREST
Accuracy	91.6%	91.0%
Precision	87.3%	84.9%
Recall	78.2%	78.2%

*The **Logistic Regression** model outperformed Random Forest in accuracy and precision. This demonstrates that the underlying relationship of the variables is relatively linear and does not require the extra complexity of ensemble models.*

Customer Profiles (K-Means)

Loyal (C1 & C2)

Characterized by their high frequency of visits, very long contracts, and higher spending than the rest.

Average (C0 & C4)

Show a more stable behavior and a completely moderate level of activity and interaction.

At Risk (C3)

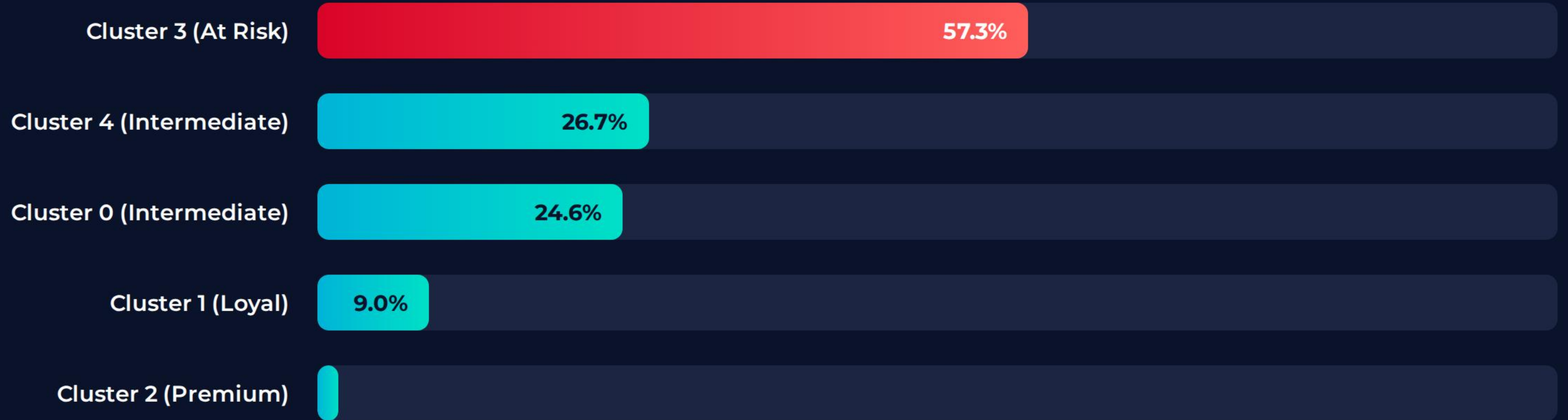
Notably low recent frequency, mostly newly registered, and with low extra spending.



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Cancellation Rate by Cluster



An indisputable pattern is observed: the risk of churn experiences a sharp drop as regular activity, system lifetime, and long-term commitments increase.

Key Conclusions



Frequency is key: A decrease or inconsistency in the number of recent visits is the behavioral metric that most accurately anticipates a user's future churn.



Power of long contracts: Encouraging extended commitments (6 to 12 months) creates a structural barrier that drastically reduces churn vs. month-to-month payments.



Complementary spending as a loyalty metric: Customer investment in peripheral services acts as an excellent barometer of satisfaction, linking to long-lasting retention.



Timely detection of the high-risk group: New subscribers who lose their initial training rhythm represent the critical cluster and require swift, personalized attention (Cluster 3).

Strategic Recommendations



Proactive Monitoring

Detect via automated alerts those users who reduce their attendance to launch quick contact campaigns and tailored promotions.



Encourage Activity

Implement innovative reward programs, new fitness challenges, and exclusive benefits to reward customer consistency and effort.



Lasting Loyalty

Redesign pricing by offering aggressive discounts on annual memberships, while promoting and facilitating social connection in group classes.